

جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
امتحان دبلوم مدارس التكنولوجيا التطبيقية الدولية الصناعية
نظام السنوات الثلاث
التجريبي ٢٠٢٥

الزمن : ساعتان

المادة : Advanced physics

التخصص : جميع التخصصات الصناعية لمدارس - WE - HST وجميع المدارس الدولية الصناعية بالحوسبة

وتكنولوجيا المعلومات (السويدي - احمد ضيف الله)

توقيع المراجع	توقيع المقدر	التقديرات			المجموعات
		R	W	NA	
					الأولى
					الثانية
					الثالثة
					الرابعة
					الخامسة
					السادسة
					السابعة
					الثامنة
					التاسعة
					العاشرة
					الحادية عشر
					الثانية عشر
					الثالثة عشر
					مجموع التقديرات

مجموع التقديرات

لا يوجد إجابة	خطأ	صح
NA	W	R

توقيع المراجع النهائي

تنبيه هام

الإجابة في نفس ورقة الأسئلة

رقم المراقبة

٢٢٠٧

وزارة التربية والتعليم والتعليم الفني
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رقم المراقبة

المادة / Advanced physics

المدرسة

الاسم واللقب

رقم الجلوس

Group 1 ;**(A) Write the scientific term expressed by the following statements:**

	Statement
1	It is a vector emanating from North Pole of the coil and perpendicular to its area)
2	It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field

(B) Put a sign (✓) if the statement is correct and an (✗) sign if the statement is incorrect:

	statement	
3	The force acting on a straight wire through which a current flows in a magnetic field decreases with increasing length of wire.	(.....)
4	The density of magnetic field produced by a solenoid carrying a current at a point on its interior axis increases when a soft iron bar inserted in it	(.....)
5	The magnetic torque is maximum value When the perpendicular to coil's plane normal to the magnetic field	(.....)
6	The magnitude of the dipole moment doesn't depend on the electric current intensity	(.....)

(C) Correct the underlined word

7	a) The rule used to determine the direction of the magnetic force is called <u>right hand screw rule</u>
8	b) The force acting on a wire of length 5 m carrying current of 20A placed in a magnetic field 0.01T is equal to <u>10N</u>

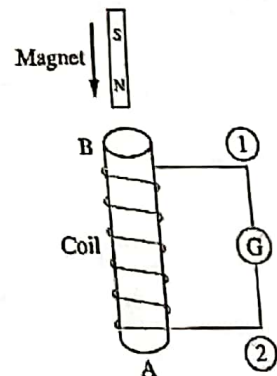
(D) Put circle around the correct answer

- 9 A solenoid carrying current has a magnetic flux density ($2B$) at a point on its axis. If it is stretched to double its length with same current intensity, then magnetic flux density at a point on its axis is:
 A) B
 B) $\frac{1}{2}B$
 C) $2B$
 D) $4B$
- 10 The magnetic flux density at a point located vertically at a distance d from a straight wire carrying a current of intensity I equals X . When increasing the current intensity to double, then the magnetic flux density becomes...
 A) $0.5X$.
 B) $2X$.
 C) X .
 D) $4X$.

Group 2 :**A) Put circle around the correct answer**

- 11 When the number of turns in the dynamo's coil is doubled, then the maximum generated electromotive force generated from it:
 a) Increases to double
 b) Decreases to its half value
 c) Remained constant
 d) Increases 4 times
- 12 A bar magnet is dropped into a solenoid connected to a galvanometer, as shown in the opposite figure. Which choice of the following is correct?

	The magnetic pole formed at A	The magnetic pole formed at B
a)	North	North
b)	South	South
c)	South	North
d)	North	South



B) Mention the mathematical formula for each of the following

13	Mutual induction coefficient	
14	average induced emf in a coil	

C) Compare between the following:

15	Fleming's right hand rule	Lenz rule
16		
Uses		

D) Write the scientific term expressed by the following statements:

17	induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect
18	The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil

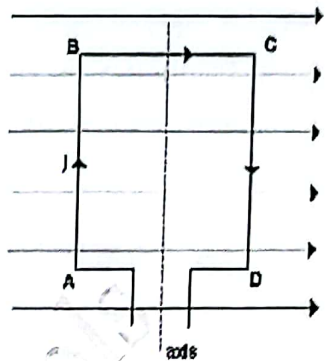
(E) Put a sign (✓) if the statement is correct and an (✗) sign if the statement is incorrect:

	statement	
19	The ratio of output electric power to the input electric power in the electric transformer is called efficiency of transformer	(.....)
20	If the wires moves by an angle along the direction of the magnetic field an induced emf generated between two ends of the wire is given by : $emf = \frac{Blv}{\sin\theta}$	(.....)

Group 3 :**A) Correct the underlined word in each of the following**

	Statement
21	A circular coil of radius 0.0157 m carries a current 2 A. The magnetic flux density at the Centre of the circular coil is $(\mu_{air} = 4\pi \times 10^{-7} \text{ wb/Amp.m})$ a) $1.57 \times 10^{-3} \text{ T}$ b) $8 \times 10^{-5} \text{ T}$ c) $2 \times 10^{-3} \text{ T}$ d) $3.14 \times 10^{-1} \text{ T}$
22	The following graph shows the relation between the magnetic flux density B at the center of four different current-carrying circular loops, 1, 2, 3, and 4, and the intensity of the current passing in each of the loops. Which of these four loops has the smallest radius? a) Loop 1 b) Loop 2 c) Loop 3 d) Loop 4
23	Which combination of core and current through the coil of wire will produce the greatest flux density along the axis of solenoid
24	In Faraday's Experiment for electro- magnetic induction, the induced emf in the coil increases when: a) Keeping the magnet still inside the coil. b) Speeding up the magnet motion relative to the coil. c) Connecting a galvanometer to the coil. d) Increasing the spacing between the coil turns.

B) In the opposite figure: complete the following.

25	Consider a rectangular coil of wire (ABCD) carrying a current (I) is placed in a magnetic field of flux density (B), with its plane parallel to the field direction, as shown in the opposite figure		
26			
	force acting on Wires (AD) and (BC)	force acting on Wires (AB) and (CD)	
			

C) Answer the following

27	Determine the magnetic flux density at a distance of 10 cm in air from the center axis a long wire carrying a current of 10A ($\mu_{\text{air}} = 4\pi \times 10^{-7} \text{ wb/Amp.m}$)
28	
	

(D) Put a sign (✓) if the statement is correct and an (X) sign if the statement is incorrect:

	statement	
29	From Lenz's rule, the direction of induced current in generated in the conductor depends on the direction of the motion. and the direction of the magnetic field.	(.....)
30	When the magnet is moved away from the solenoid, magnetic flux link with the coil increases, the coil's near end forms like pole which tends to repel it	(.....)

Group 4 :

A) Correct the underlined word in each of the following

31	There is an induced emf in the secondary coil only when The current in the primary coil is <u>constant</u> .
32	Switching on the current in primary coil or increasing the current intensity in primary coil by using rheostat. <u>producing Forward induced emf</u>
33	<u>Alternating current (AC)</u> has <u>constant</u> direction and its intensity periodically changed
34	The ac generator is a device used for converting <u>heat</u> energy into electrical energy.

B) Choose the correct answer

B) Choose the correct answer

- 35 A rectangular coil of 100 turns and cross-sectional area 0.06 m^2 rotates at
36 frequency 50 Hz in a uniform magnetic field of flux density 0.1 T.
The maximum induced emf is
- (A) 30 V
(B) 94.2 V
(C) 188.5 V
(D) 60 V

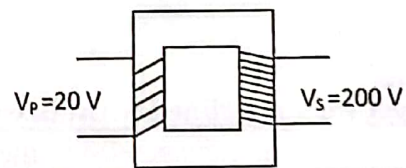
C) Compare between each of the following

37	Point of	ac generator	transformer
38	comparison		
	working principle		

- 39 Mention the shape of magnetic field due to a current in a straight wire:
40

Group 5: Answer the following

- 41 Is this step-up or step-down transformer? Why?
42



43
44

In which figure magnetic flux equal zero and in which of them equal maximum?

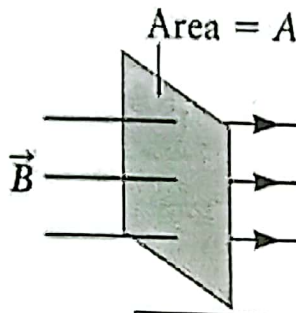


Figure (1)

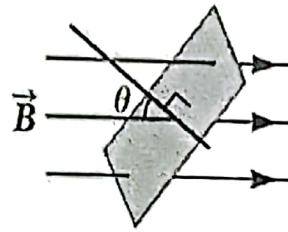


Figure (2)

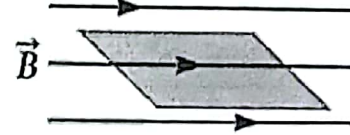


Figure (3)

45
46

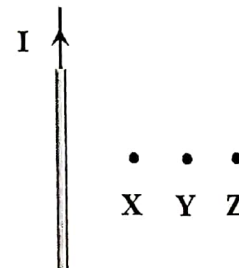
A rectangular loop of length 0.08 m, width 0.05 m and 100 turns is placed in a magnetic field of 10 Tesla flux density, carrying current 1 A, Find The maximum torque acting on the coil

47
48

mention two Applications on which the concept of the couple (magnetic torque) acting on the coil carrying current placed in a magnetic field is applied

49
50

A long straight wire carrying electric current of intensity (I) as in the figure. Which one from the following points (x, y, z) has maximum the magnetic flux density (B) ?



(انتهت الاسئلة)

Answer the following groups

Group 1 :

(A) Write the scientific term expressed by the following statements:

	Statement
1	dipole moment
2	Tesla

(B) Put a sign (✓) if the statement is correct and an (X) sign if the statement is incorrect:

	statement	
3		(x)
4		(✓)
5		(✓)
6		(x)

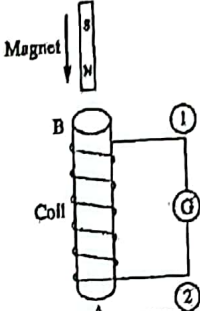
(C) Correct the underlined word

7	Fleming left hand rule
8	2N

D) Put circle around the correct answer

<u>9</u>	B
<u>10</u>	2X

Group 2 :**A) Put circle around the correct answer**

11	Increases to double								
12	<table border="1"><thead><tr><th></th><th>The magnetic pole formed at A</th><th>The magnetic pole formed at B</th></tr></thead><tbody><tr><td>c)</td><td>South</td><td>North</td></tr></tbody></table> 				The magnetic pole formed at A	The magnetic pole formed at B	c)	South	North
	The magnetic pole formed at A	The magnetic pole formed at B							
c)	South	North							

B) Mention the mathematical formula for each of the following

13	Mutual induction coefficient	$emf = M \frac{\Delta I}{\Delta t}$
14	average induced emf in a coil	$emf = N \frac{\Delta \phi}{\Delta t}$

C) Compare between the following:

15	Fleming's right hand rule	Lenz rule
16		
Uses	determine the direction of induced current in straight wire moving in a magnetic field	determine the direction of induced current in a coil

D) Write the scientific term expressed by the following statements:

17	Eddy current
18	self-induction

(E) Put a sign (✓) if the statement is correct and an (X) sign if the statement is incorrect:

19	(.....✓....)
20	(.....X....)

Group 3 :**A) Correct the underlined word in each of the following**

Statement	
21	$8 \times 10^{-5} \text{ T}$
22	Loop 1
23	B
24	Speeding up the magnet motion relative to the coil.

B) In the opposite figure: complete the following.

25	force acting on Wires (AD) and (BC)	force acting on Wires (AB) and (CD)
26	Zero	max.

C) Answer the following

27	$B = 2 \times 10^{-7} \times \frac{10}{0.1} = 2 \times 10^{-5} \text{ T}$
28	

(D) Put a sign ($\checkmark$) if the statement is correct and an (X) sign if the statement is incorrect:

29	(..... $\checkmark$)
30	(.....X.....)

Group 4 :**A) Correct the underlined word in each of the following**

31	variable
32	backward
33	variable
34	mechanical

Choose the correct answer

35	(C) 188.5 V
36	

Compare between each of the following

37	Point of comparison	ac generator	transformer
38	working principle	electromagnetic induction	mutual induction

circular magnetic flux lines

Group 5: Answer the following

41	
42	Step up because number of turns of secondary coil or voltage is larger
43	
44	Figure 1 Figure 3
45	
46	$\text{Torque} = BIAN = 10 \times 1 \times 100 \times 0.08 \times 0.05 = 4 \text{N.m}$
47	
48	Motor and galvanometer
49	
50	Point X.

(انتهت الاجابة)